10. Exploratory Data Analysis (EDA)

Numeric overview

Field | Non-null | Mean | Median | Min | Max

battery_capacity_kwh | 100,000 | 78.23 | 63.30 | 1.50 | 410.00

battery_cycle_life | 97,972 | 2,072.45 | 2,025.00 | 500.00 | 4,375.00

battery_degradation_pct_5yr | 100,000 | 12.29 | 12.30 | 2.00 | 36.14

range_km | 100,000 | 324.31 | 338.30 | 55.00 | 659.90

efficiency_wh_per_km | 100,000 | 250.57 | 165.18 | 9.98 | 2,242.84

motor_power_kw | 100,000 | 169.86 | 140.20 | 1.50 | 750.00

max_dc_charging_kw | 100,000 | 152.23 | 138.60 | 3.00 | 350.00

price_usd | 100,000 | 65,010.46 | 42,808.69 | 900.04 | 409,995.49

annual_sales_units | 99,011 | 7,803.26 | 3,935.00 | 10.00 | 772,442.00

customer_rating | 98,069 | 4.08 | 4.09 | 1.75 | 5.00

co2_saved_tons_per_year | 100,000 | 3.69 | 3.00 | 0.95 | 13.30

Category composition

The most frequent category is **Two-Wheeler Scooter** with 9,826 records (9.83%). The dataset contains 15 categories. The largest region by record count is **North America** (14,427 records).

Model-year coverage

Model year | Records | Mean price | Mean range | Mean customer rating

2018 | 11127 | $64,954.16 | 323.3 km | 4.09

2019 | 11043 | $66,072.36 | 324.9 km | 4.09

2020 | 11029 | $64,522.57 | 325.1 km | 4.09

2021 | 11172 | $64,915.89 | 325.0 km | 4.08

2022 | 11174 | $65,433.84 | 324.1 km | 4.09

2023 | 11060 | $64,653.86 | 323.3 km | 4.08

2024 | 11213 | $64,638.64 | 322.3 km | 4.08

2025 | 10910 | $65,097.92 | 325.6 km | 4.08

2026 | 11272 | $64,812.24 | 325.1 km | 4.08

Numeric relationships

Field A | Field B | Pearson r

price_usd | battery_capacity_kwh | 0.837

price_usd | range_km | 0.277

battery_capacity_kwh | efficiency_wh_per_km | 0.908

range_km | motor_power_kw | 0.602

annual_sales_units | price_usd | -0.117

customer_rating | price_usd | -0.001

EDA interpretation

- Mean price is $65,010.46, while median price is $42,808.69; the gap indicates a right-skewed price distribution.
- Mean range is 324.3 km and median range is 338.3 km.
- Price and battery capacity have a strong positive correlation in this file (r = 0.837); this is association, not proof that capacity alone determines price.
- Customer rating has near-zero linear correlation with price (r = -0.001) in this dataset. This does not prove ratings are independent of all vehicle characteristics.
- Annual sales contain a very high maximum (772,442) compared with the median (3,935); robust summaries and record-level review are important.

Suggested charts

1. Count of records by category.
2. Median and mean price by category.
3. Range distribution by category.
4. Battery capacity vs range scatterplot.
5. Price vs battery capacity scatterplot.
6. Annual sales distribution with log scale or percentiles if justified.
7. Region and model-year counts.